MIL-S-2438C
31 January 1973
SUPERSEDING
MIL-S-2438B(MC)
7 April 1967

MILITARY SPECIFICATION

SCREEN, HALFTONE: CROSSLINE, WITH CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type rectangular, glass, crossline pattern, halftone screen.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-F-336 - Film, Photographic, Photomechanical, Black-and-White.
PPP-B-601 - Box, Wood, Cleated-Plywood.
PPP-B-621 - Box, Wood, Nailed and Lock Corner.
PPP-B-636 - Box, Fiberboard

MILITARY

MIL-P-116 - Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

FSC 3610

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. The first article is defined as a preproduction sample which will be identical to the screen the manufacturer will subsequently produce in fulfillment of the contract. Unless otherwise specified (see 6.2), as soon as practicable after award of the contract or order and prior to submission of any complete items for final acceptance, the contractor shall furnish a first article for testing to determine conformance with this specification. Approval of the first article by the activity concerned shall not relieve the contractor of his obligation to supply halftone screens conforming to this specification. Any changes or deviations from the first article shall be approved by the contracting officer.

* 3.2 Materials. Materials shall be as specified herein. Material not definitely specified shall be of a good commercial quality and entirely suitable for the intended purpose. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

* 3.3 Design and construction. The halftone screen shall be essentially the standard product of the manufacturer so that prompt and continuing service may be assured. The screen shall consist of two glass plates, with a crossline pattern between them, joined together with optical cement and provided with a metal binding around the periphery.

* 3.3.1 Glass plates. The glass plates shall be especially selected polished plate glass of the highest quality. The glass plates shall be free from scratches and striae. No bubbles shall be permitted in screens up to 16 inches in nominal width. Not more than one bubble in any area 10 inches square shall be permitted in screens over 16 inches in nominal width. Bubbles shall be not larger than twice the width of the screen apertures (see 3.3.3). Surfaces of the glass plate shall be free from waves and shall be flat and parallel to each other within one minute of arc (see 4.5.2).

* 3.3.2 Cover glass. The physical thickness of the cover glass plate shall be not more than 0.120 inch for screens having a nominal width less than 26 inches, and shall be not more than 0.140 inch for screens having a nominal width of 26 inches or greater. The optical thickness of the cover glass shall be equal to the physical thickness marked on the edge of the screen divided by the refractive index of the glass, within ± 1 percent when tested as specified in 4.5.3. When the refractive index is not known accurately, a value of 1,500 shall be assumed.

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* 3.3.3 Screen pattern. The screen pattern shall consist of transparent square apertures on an opaque background, such as that produced by two series of parallel lines crossed at right angles. The background shall be essentially black on both sides of the screen. The entire area enclosed within the metal binding shall appear uniformly gray, with no irregularities of any kind, when inspected as specified in 4.5.4. All apertures shall appear transparent and all background areas shall appear opaque when examined with a microscope (see 6.3). All aperture edges shall be straight and sharply defined, and shall lie essentially in the same plane. All aperture corners shall be square, or alternatively, all corners shall be rounded with approximately the same radius, not to exceed 5 percent of the aperture width. The sum of the aperture width and the distance between apertures shall be 1/(screen ruling), ±2 percent (see 3.3.3). The ratio of the distance between aperture width shall be 1.00, +0.02 -0.01. The angles between adjacent sides of all apertures shall be 90°, ± one-fourth of a degree. The angles between aperture sides and the outer edges of the metal binding shall be 45°, ± one-fourth of a degree.

* 3.3.3.1 Screen ruling. The screen ruling, defined as the number of apertures per inch along a line parallel to the aperture sides, shall be as specified (see 6.2), ±2 percent.

* 3.3.3.2 Light scattering. When examined as specified in 4.5.5, the edges of the apertures shall appear as uniformly narrow, continuous lines of light, forming complete square outlines. There shall be no other light spots or lines.

* 3.3.4 Optical cement. The cement used to join the glass plates shall be the best grade of optical cement. The cement shall have approximately the same index of refraction as the glass, and shall be optically homogenous and noncrystallizing. The cement shall not deteriorate, opalesce, or darken under the temperature conditions specified in 3.4. There shall be no bubbles, blisters, lint, or dust particles in the cement within the nominal area of the screen. The thickness of the cement shall be such that a metallic ring and not a dull thud will result from striking the screen, when tested as specified in 4.6.6.

* 3.3.5 Edge binding. The edge binding shall be of copper, aluminum, nickel, or alloys therof. The binding shall be so fitted to the glass that water, alcohol, or other liquids cannot reach the optical cement at the edges of the glass plates. The binding shall consist of a preformed channel strip fitted and cemented to the glass, or shall be electrodeposited directly onto the edges of the glass. The surface of the binding shall be flush with or below the glass surface on both sides of the screen. When tested as specified in 4.5.6, no light shall appear between the straight edge and glass surface. The angle between adjacent outer edges of the binding shall be 90°, ±10 minutes.

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* 3.3.6 Screen case. Each screen shall be provided with a plush lined, wooden case of the type usually supplied by the manufacturer to commercial users. The case shall be finished in accordance with good commercial practice.

* 3.4 Performance. The screen shall produce halftone tint negatives having a uniform density of 0.57, +0.05 or -0.04 corresponding to a calculated black dot area of 70 percent, ±3 percent when tested as specified in 4.6.1. The tint negative produced shall be at least as large as the nominal screen size. The tint negative shall have no spots, lines, or mottles visible to the unaided eye, except those caused by the distorted shape or size of one screen aperture, or by one bubble in the glass (see 3.3.1). These defects shall be limited to not more than two in any area 10 inches square. The screens shall be capable of satisfactory performance in any ambient temperature from +125°F to +50°F, and shall not be damaged by storage in any ambient temperature from +160°F, for periods of 4 hours daily to -80°F, for periods of several days at a time.

* 3.5 Dimensions. The nominal size of the screen shall be as specified (see 6.2). The screen shall cover an image having the nominal screen size. The overall dimensions and tolerances of the completed screens shall conform to Table I.

Table I. Dimensional tolerances

	Nominal screen dimensions		
	under 26 inches	26 to 36 inches	over 36 inches
	Inch	Inch	Inch
Maximum difference between overall and nominal widths and lengths.	0.625	0.750	0.750
Overall thickness			
Minimum	0.200	0.200	0.200
Maximum	0.260	0.275	0.275
Maximum variation in thickness of a single screen	0.010	0.012	0.015
Maximum deviation from a plane of outer cover glass surface and of screen pattern, when positioned in a vertical plane and free from external twisting forces.	±0.005	±0.007	±0.015

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* 3.6 Identification marking. The screen shall be marked for identification in accordance with MIL-STD-130. "Special Characteristics" shall include the following information:

(a) Nominal size, in inches.
(b) Screen ruling, lines per inch.
(c) Physical thickness of the cover glass (as a decimal fraction of an inch, to the nearest 0.001 inch).

All identification marking shall be etched or engraved either on the cover glass side of the metal binding or on the surface of the cover glass outside the nominal size.

* 3.7 Workmanship. Workmanship shall be of the highest quality throughout and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Quality conformance (see 4.4).

(c) Preparation for delivery (see 4.7).

* 4.3 First article inspection. The first article screen, selected in accordance with 3.1, shall be subjected to examinations in 4.5 and the tests in 4.6. Unless otherwise specified in the contract or order (see 6.2), testing of the first article shall be performed at the place of manufacture.

* 4.4 Quality conformance inspection. The quality conformance inspection shall consist of examinations in 4.5 and tests in 4.6. Each screen shall be visually and dimensionally examined and tested to determine conformance with this specification.

* 4.5 Examination. The examination shall be conducted as follows:

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* 4.5.1 Uniformity of pattern. The screen shall be examined by transmitted light against a uniformly illuminated white background, using the unaided eyes, for conformance with 3.3.3.

* 4.5.2 Glass surfaces. Both outer surfaces of the finished screen shall be visually inspected by reflected light with the unaided eyes for conformance with 3.3.1.

* 4.5.3 Cover glass thickness. The cover glass thickness shall be measured at two widely separated points on the completed screen by means of a microscope (see 6.3), using the calibrated fine adjustment drum to determine conformance with 3.3.2. The microscope shall be placed on the cover glass side of the screen (the side having the identification markings) and focused first on the outer surface of the cover glass, either on dust particles or on a mark made with a glass marking pencil. The microscope shall then be focused on the screen pattern, using only the fine adjustment drum. The difference between readings on the fine adjustment drum is the optical thickness, or equivalent air thickness, of the cover glass, in microns (1 micron = 0.00003937 inch).

* 4.5.4 Screen pattern. The screen pattern shall be inspected at no less than 20 places distributed uniformly over the screen area, using a microscope (see 6.3), for conformance with 3.3.3. The aperture width and distance between apertures, in two directions, shall be measured at not less than 10 places distributed uniformly over the screen area. The apertures shall then be examined, using a single focus setting of the microscope at not less than 20 places distributed uniformly over the screen area, to determine whether aperture edges lie essentially in the same plane and whether the cover glass thickness is essentially uniform.

* 4.5.5 Dark-field inspection. The screen pattern shall be examined with the microscope by dark-field illumination at no less than 20 places distributed uniformly over the screen area to determine conformance with 3.3.3.2. Dark-field illumination shall be provided by placing a light source on the side of the screen opposite to that of the microscope, but off the microscope axis, so that the area under observation is illuminated by light falling on it at an angle so that both the apertures and the black portions of the screen pattern appear black.

* 4.5.6 Edge binding. The position of the surface of the edge binding with respect to the glass surfaces shall be determined at all points around both sides of the screen to determine conformance with 3.3.5. A machinist's straightedge shall be placed on the glass surface, extending over the edge binding and at approximately right angles to it. A light shall be placed on one side of the straightedge and observed from the other side.

* 4.6 Tests. Tests shall be as follows:

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* 4.6.1 Camera conditions. A process camera shall be used in testing the screen. There shall be no light leaks in or around the bellows, and no reflecting surfaces at any location inside the camera to reflect light from the lens position to any point in the plane of the film. The lens carriage shall be adjusted as in focusing until the distance from the inner surface of the lens board to the film plane is equal to or greater than 1.5 times the length of the diagonal of the screen to be tested. The lens and lens board shall then be removed from the camera and a specially made aperture plate substituted in which a square stop aperture has been cut. The sides of this aperture shall be vertical and horizontal (for 45° screen angle) and their lengths equal to 1/45.25 times the distance from aperture to film plane. A sheet of flashed opal glass shall be mounted over the stop aperture on the side opposite to that of the film. The inner surface of the stop aperture plate shall be finished with dull black lacquer or paint. A light score shall be located outside the opal glass so as to illuminate the aperture area uniformly.

* 4.6.2 Screen distance. The halftone screen under test shall be mounted in the camera, cover glass side facing the stop aperture, and the screen distance set at 64 times the width of the screen apertures specified in 3.3.3. In setting this distance, the spacing between the film plane and the cover glass surface shall be adjusted to the difference between the required screen distance and the optical thickness of the cover glass. The optical thickness of the cover glass shall be taken as two-thirds of the physical thickness marked on the margin of the screen.

* 4.6.3 Exposure. A sheet of film conforming to L-F-336, of sufficient size to cover the entire screen area, shall be mounted in the camera and an exposure made through the screen. The exposure shall be found by trial in order that finished negatives meeting the following requirements shall be formed. Using a magnifier at approximately 14 diameters magnification, the black areas shall be opaque, the transparent areas clear and free from fog, the borders sharp, and the black dots shall have a fractional area of approximately 70 percent by estimation. The transmission density of the negative shall then be measured at several points by means of a transmission densitometer suitable for reading halftone densities. The density shall be as specified in 3.4. If the tint negative does not meet the requirements described above, further negatives shall be made by the same procedure until negatives are obtained which do meet the requirements and are representative of all portions of the screen area.

* 4.6.4 Processing. The exposed film shall be developed in the type of developer and at the temperature recommended by the film manufacturer. Developing shall be performed in a tray. The developer shall cover the film at all points to a depth of not less than one-fourth of an inch. During the entire time of development, the developer shall be continuously brushed over the emulsion with alternating lateral and to-and-fro strokes, using a soft camel hair brush. The width of the brush and the speed of brushing shall be such that uniform developer action takes place at all points on the emulsion. Rocking of the tray will not

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provide the desired uniformity of action. The time of development may be adjusted to yield the dot size specified in 4.6.3 but must fall within the limits recommended by the film manufacturer. The developed film shall then be quickly immersed in a stop bath, agitated during immersion and then fixed, washed, and dried.

* 4.6.5 Visual examination. The one or more tint negatives selected as meeting the requirements in 4.6.3 shall then be inspected with the unaided eyes by transmitted light against a uniformly illuminated white background for uniformity of tone and for spots, lines, or mottles, either lighter or darker than surrounding areas. Judgement shall be limited to areas completely surrounded by other areas having a density of 0.53 to 0.62. Where local irregularities are found in one or more tint negatives, the actual causes shall be determined. When found definitely to be caused by some imperfection in the screen, other than those permitted in 3.4, and not some nonuniformity in the camera, photographic emulsion, exposure, or processing, the screen shall be rejected.

* 4.6.6 Thickness of optical cement. The thickness of the optical cement shall be tested for conformance with 3.3.4 by holding the completed screen in one hand, the metal binding resting on the fingers and one face touching the thumb, and striking the glass surface with a knuckle of the other hand.

* 4.7 Inspection of preparation for delivery. An inspection shall be made to determine compliance with packaging, packing, and marking requirements in Section 5. The defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery, and selected just prior to the closing operation. Sampling shall be in conformance to MIL-STD-105. The lot size shall be the number of shipping containers to be shipped at one time, to one destination, on one contract or order. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

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Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible; of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners or container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers, inadequate securing or sealing of wrapping materials, or loose wire ties.
Weight	Gross or net weight exceeds requirement.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Each screen and screen case shall be cleaned in accordance with process C-1, of MIL-P-116.

5.1.1.2 Drying. Each screen and screen case shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packaging. Each screen within screen case shall be packaged in accordance with method IC-2, of MIL-P-116.

5.1.2 Level C. Each screen within screen case shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. A quantity of screens within screen cases, packaged as specified in 5.1, shall be packed in close-fitting cleated-plywood, nailed wood or fiberboard boxes conforming to PPP-B-601, overseas type, style A or B, PPP-B-621, class 2, style 4 or PPP-B-636, type CF, class weather-resistant, special requirements use. Shipping containers shall be closed, strapped and reinforced in accordance with the appendix to the applicable box specification. The gross weight of wood type shipping containers shall not exceed 200 pounds. The gross weight of fiberboard shipping containers shall not exceed the special requirements limitations of Table II of PPP-B-636.

5.2.2 Level B. A quantity of screens within screen cases, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that wood type shipping containers shall be domestic type and class 1, respectively, and fiberboard containers shall be class domestic.

5.2.3 Level C. A quantity of screens within screen cases, packaged as specified in 5.1, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use. The screens are used to make halftone negatives and positives for offset lithography or letterpress printing.

* 6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When first article is not required (see 3.1).
(c) Screen ruling, lines per inch (see 3.3.3.1).
(d) Size of screen required (see 3.5).
(e) Where first article will be tested if elsewhere than as indicated in 4.3.
(f) Level of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 Microscope. The microscope used shall provide an overall magnification of approximately 75 diameters. The microscope shall be equipped with an eyepiece of known calibration or other device for measuring distances in a plane normal to the axis of the microscope, and shall have a calibrated focusing drum for measuring vertical distances.

* 6.4 First article. First article examination, test and approval shall be as specified by the contracting officer (see 3.1).

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* 6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy - MC
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Navy - MC
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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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